

ClaimsForge

Multi-agent AI for automated e-commerce damage claims —
Gemini Vision evidence + policy reasoning + tone-aware
response, end-to-end in 8 seconds.

Try it live

<http://45.32.154.255>

github.com/zhenyueD/claimsforge · MIT



E-commerce loses \$743B/year to returns.

\$743B

Global e-comm returns / year

**2-4
days**

Avg time to resolve a damage claim

15 min

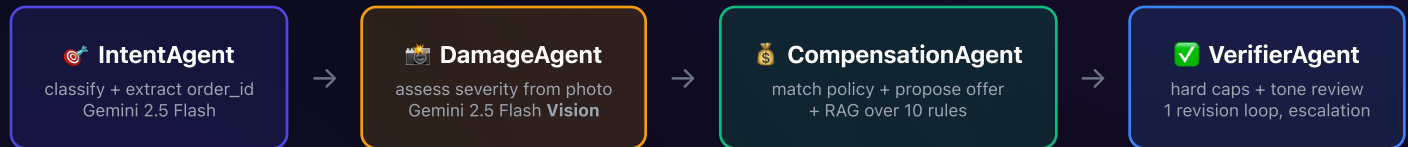
Human agent time per low-value claim

73%

Of claims are <\$50 (auto-resolvable)

Customer support agents still read photos and argue over \$20 refunds. **This is solved.**

4 specialist agents · 1 pipeline · 8 seconds



Plain Python orchestrator (~150 LOC). No LangChain. Streams agent_trace events to WebSocket for live UI.



DamageAgent reads evidence.

Input

- User text: "我的马克杯杯口裂了 3cm"
- Image bytes (auto-resized to 1024px)
- Structured output schema (Pydantic)

Output

```
{
  "damage_type": "crack",
  "severity": 8,
  "affected_parts": ["杯口"],
  "confidence": 0.90,
  "reasoning": "图片显示杯口
  有明显裂纹, 与用户描述
  的3cm裂纹一致..."
}
```

Native Gemini features used: `response_schema`, `thinking_budget=0`, `multimodal` `Part.from_bytes()`.



CompensationAgent cites its rules.

10 policies (data/policies.json)

- P-RET-01 severity ≥ 7 → full refund (cap ¥500)
- P-RET-02 severity 4-6 → 50% refund
- P-RET-04 易碎品 + crack ≥ 5 → replacement
- P-RET-05 电子产品进水 → **force escalate**
- P-EMO-01 emotion ≥ 8 → +20% uplift
- P-LMT-01 no image + $\geq$ ¥50 → escalate

Hybrid filter → LLM proposal

Deterministic filter narrows policies by severity / type / emotion. Gemini picks the best fit and writes an empathetic justification:

"我们很抱歉您收到的马克杯有裂纹。考虑到商品损坏情况，我们将为您办理全额退款 ¥24.00..."

Built on the latest Gemini features.

Native features

- `response_schema` + Pydantic — zero JSON parsing
- `thinking_budget` control per agent
- Multimodal in one call (image + text + schema)
- Function calling for tool-using agents
- `structured()` wrapper for safe typed outputs

Stack

FastAPI

Python 3.12

Pydantic 2

google-genai 2.x

Pillow

WebSocket (live trace)

Docker

Vultr Cloud Compute

nginx + Let's Encrypt

~600 LOC new code. Built on the proven easyclaw-demo backbone (RAG, emotion analyzer, escalation engine).

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ClaimsForge — Milan AI Week 2026

Real numbers per 1,000 low-value claims:

90%

Auto-resolved (vs 5% today)

12s

Avg resolution time (vs 2-4 days)

\$3,750

Agent-hour savings (@ \$15/hr)

+18 pt

Projected CSAT lift

Who pays for this?

- Mid-market e-commerce (10k+ orders/month) — ROI in <30 days
- Returns aggregators (Loop, Returnly, Happy Returns) as a feature
- Marketplaces (Etsy, Mercari) selling it to their sellers

ClaimsForge

Multi-agent Gemini-powered claims resolution. Open source MIT. Deployed on Vultr.

Live

<http://45.32.154.255>



Code

github.com/zhenyueD/claimsforge